

Dataset-to-Simulation Mapping: Governing Energy Functional

$$\Pi(\mathbf{u}, d) = \int_{\Omega} [g(d)\Psi_0^+ + \Psi_0^-] d\Omega + \int_{\Omega} G_{c,b} \gamma(d, \nabla d) d\Omega + \int_{\Gamma_c} \phi(\Delta_n, \Delta_t) d\Gamma$$

Elastic Energy (UMAT)

$E(T), \nu(T) \rightarrow$ Sec. 2

$\alpha(T)\Delta T \rightarrow$ Sec. 2

$\beta\Delta\delta \rightarrow$ Sec. 3

Phase-Field Fracture

$G_{c,b}(T) \rightarrow$ Sec. 4

$l_0 \rightarrow$ Sec. 1 (mesh)

$g(d) = (1 - d)^2$

Cohesive Interface (UEL)

$G_{c,I}, G_{c,II} \rightarrow$ Sec. 4

$T_{max,n}, T_{max,t} \rightarrow$ Sec. 4

$\eta_{BK} \rightarrow$ Sec. 4

Experimental Validation (Sec. 5)

$\kappa(T)$ curvature | Crack path overlay | Delamination onset ($T_c, p_{O_2,c}$)